

A Review of A Multivariate Approach to the Study of Orichalcum ingots from the Underwater Gela's Archaeological Site

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1. Abstract

General Background

A shipwreck off the coast of Gela, Italy has been the source of a historical find of 40 orichalcum ingots. A scarce, fabled metal that has the modern equivalent of brass.

sPECIFIC BACKGROUND

Knowledge gap

Large groupings and finds from one source are very hard to come by: lots of data in one place.

Here we show...

After performing a trilogy of cluster analyses, we show that were the clusters accurate were we able to determine a good fit?

Our goal is to create and present a comprehensive, comprehensible, reproducible, and replicable multivariate analysis of the data obtained from the chemometric elemental analysis of 38 orichalcum ingots that were discovered in a 6th Century BC shipwreck off the coast Gela, Italy in 2015.

Results

Implications

They had different origins. Can possibly be linked to geological locations by ore analysis, collected from different regions, etc.

Why did they wait to do further testing that would provide more information on the composition and provide more accurate data?

The groupings from the Caponetti, Armetta, Brusca, et al. (2017) study appears to have been manipulated without compensation of the . By taking the log at any point in this anylysis without standardisation would have resulted in multiple zero-values and artificially inflating

Is the cluster analysis consistent with different distances and linkages?

This is the largest sample of its kind however it is still not a large sample.

For the analysis of our report we will be considering all the main components and trace element concentrations excluding the Cu/Zn ratio as...

Why do we care about the problem and the results?

Because it could lead to further impressive archaeological finds

What is the scope of the work and what problem is this paper trying to solve?

Determine whether or not any of the ingots could be grouped. The scope is mapping orichalcum origin.

What was done to solve this problem?

More destructive testing to determine trace elements.

What implications does the answer imply?

one of our hypotheses questions is whether our data follows a multivariate normal distribution. probably some hypotheses around the clusterings.

2. Introduction

2.1 Significance of this archaeological find and Gela

The historic Gela, Italy has been a treasure trove of archaeological finds for generations. This ancient city was strategically positioned at the crux of major maritime trade routes, which made it a populous and successful trading port (Angelucci, 2026). Consequently, ships looking to trade goods like pottery, metals, and grain have traveled to and from Gela for centuries.

A shipwreck off the coast continues to provide significant archaeological finds to this day. This discovery dates back to 1988 (Nutter, 2025), where it was first stumbled upon by divers near Bulala beach. We are exploring one of the most significant finds from this wreck; A veritable hoard of the fabled metal, Orichalcum.

The discovery of 40 orichalcum ingots in 2017 was a monumental one. There had never been a discovery of this quantity in one place anywhere in the world. There was also evidence of consistent casting processes, with overall similarities suggesting a rudimentary mold production. This was likely a hole dug in the ground and molten alloy was poured in the indentation to solidify. The morphology of the found ingots suggested cast-diverse technologies.

2.2 What is orichalcum?

Orichalcum is defined as a copper alloy with zinc content as the principal components. A widely varying quantity of lead and tin among other trace elements are expected. There is no consistent proportion to the amount of copper and zinc found, but the overall composition and relative amounts of principal elemental components can be used to deduce the rough date and origin of a sample. The modern day equivalent is brass.

2.3 Summary of Collection Techniques

In the initial analysis of the orichalcum ingots, X-Ray Fluorescence (XRF) spectroscopy via a portable spectrometer (Caponetti, Armetta, Martino, et al., 2017) was used. The subject paper includes that data and further investigates the composition of each ingot using more invasive and refined elemental analysis.

To obtain material for further testing, a micro-well was drilled into each ingot and the extracted alloy was taken to determine trace elements using Inductively coupled plasma optical emission spectroscopy (ICP-OES) and Inductively coupled plasma optical emission spectroscopy (ICP-MS) of each of the 38 orichalcum ingots. (FLESH OUT AND EXPLAIN MORE, TOO VAGUE NOT EXPLAINING WHY)

2.4 Scope of This Multivariate Analysis

This multivariate analysis includes manipulations of the data collected from the ICP-OES and ICP-MS chemometric analysis, it does not include the data about physical characteristics of the ingots or any correlation analysis of those variables.

We will be excluding the Cu/Zn ratio variable as we chose to analyse the linkage of all elements between each other, not between Cu/Zn ratios and other elements

3. EDA

3.1 Analysing raw data

We performed a simple summary of the data in table 1 and creating a box plot to visualize the data. Something to note is that the elements Cu, Zn, Pb, and Fe are all in percentages, where as the concentrations of the other elements are in PPM (parts per a million). This immediately indicated that scaling was needed.

Table 1: Figure 1: Main component and trace element concentrations obtained by ICP-OES & ICP-MS techniques

Sam-																						
ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba	
S1	73	21	5.60	0.30	580	146.0	5.7	9.85	2.79	12.2	9.70	0.57	41.4	19.9	5.55	11.50	3.71	3.17	0.08	1.00	0.19	
S2	87	16	4.29	0.53	534	199.0	13.2	0.36	408.44	20.0	4.80	1.00	72.8	20.8	2.17	4.46	1.55	1.31	0.07	0.15	0.18	
S3	73	28	5.91	0.78	451	131.0	105.0	23.8	0.8	47	16.5	6.70	0.64	29.0	15.7	4.00	8.89	3.33	2.70	0.14	8.90	0.35
S4	80	20	6.17	0.39	558	171.0	70.0	11.1	04.83	11.9	9.34	0.82	60.7	23.2	6.16	13.80	6.08	4.91	0.14	0.29	0.39	
S5	78	22	3.87	0.55	513	156.0	67.0	14.5	06.49	5.3	7.69	0.58	58.7	23.3	3.87	11.60	4.38	3.69	0.12	0.91	0.45	
S6	81	22	6.51	0.19	579	159.0	60.3	8.32	1.88	9.6	8.54	0.96	61.4	15.7	2.12	4.36	2.29	1.90	0.11	0.45	0.75	

Sam-

ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
S7	75	20	6.45	0.33	559	130.0	07.0	211.0	0.37	11.6	17.6	0.80	51.0	22.3	1.75	5.50	2.16	1.57	0.13	0.34	0.50
S8	82	24	6.13	0.31	548	160.0	75.4	9.66	3.19	7.6	6.98	0.83	58.5	23.7	1.99	13.3	0.55	2.36	0.11	1.45	0.26
S9	80	18	4.63	0.44	482	159.0	09.0	24.2	05.28	30.5	3.92	0.91	91.3	23.7	1.69	8.40	2.70	1.78	0.07	0.11	0.43
S10	72	28	5.92	0.76	435	105.0	11.0	23.5	08.29	16.2	6.65	0.61	27.6	19.5	1.23	6.77	2.00	1.41	0.12	0.21	0.30
S11	76	19	5.93	0.43	525	138.0	3.2	15.1	05.13	8.5	8.29	0.86	54.5	12.9	0.92	6.57	1.71	1.28	0.10	0.56	0.24
S12	69	17	6.67	0.50	657	141.0	0.1	13.8	04.45	8.3	8.14	0.87	55.0	13.4	1.20	3.22	1.65	1.10	0.10	3.13	0.22
S13	78	20	3.30	1.06	458	189.0	49.0	97.0	0.60	17.3	38.3	0.00	181.0	86.6	1.55	9.28	3.08	2.23	0.07	0.52	1.17
S14	74	25	4.70	0.44	433	81.7	88.2	14.6	05.08	20.3	1.98	0.71	26.1	29.7	1.13	6.51	2.49	1.55	0.10	0.17	0.19
S15	84	20	5.20	0.14	549	171.0	07.0	20.6	01.06	23.4	4.24	0.95	92.0	12.0	0.74	3.44	1.58	1.08	0.08	0.08	0.17
S16	82	19	4.72	0.16	560	162.0	03.0	9.90	1.14	22.2	4.00	0.89	82.1	12.4	0.67	21.2	0.32	2.53	0.08	0.31	0.31
S17	77	20	6.92	0.79	500	177.0	153.0	4.70	9.06	31.8	65.8	0.81	27.9	5.8	0.40	3.31	0.93	0.67	0.08	7.27	0.35
S18	83	19	4.57	0.44	515	158.0	11.0	22.1	05.09	28.4	3.80	0.99	89.3	21.5	0.52	4.08	1.69	1.01	0.09	11.80	0.27
S19	71	28	5.25	0.72	494	114.0	01.0	8.8	106.14	18.5	10.40	0.72	33.6	17.7	1.49	8.36	2.61	1.54	0.16	4.89	3.37
S20	76	19	6.31	0.41	535	139.0	1.5	14.0	04.41	8.2	8.11	0.97	56.3	13.4	0.62	2.91	1.34	0.79	0.06	0.42	0.40
S21	74	21	4.88	0.49	423	80.0	68.0	14.4	05.67	22.5	1.99	0.91	25.3	26.3	0.42	2.49	0.97	0.59	0.07	0.70	0.23
S22	71	21	6.00	0.93	606	142.0	175.0	5.30	15.03	11.8	42.00	0.56	46.3	25.7	0.85	11.6	0.82	2.19	0.12	15.10	0.61
S23	72	22	6.18	0.33	623	121.0	67.2	9.12	2.61	11.6	9.07	0.58	40.1	18.3	0.73	5.38	2.52	1.45	0.11	0.08	0.14
S24	76	18	6.49	0.28	645	172.0	52.5	9.43	1.94	9.1	9.75	1.00	79.7	14.6	0.40	3.76	1.39	0.86	0.08	0.15	0.28
S25	78	18	1.05	0.75	496	162.0	2.2	56.7	00.34	5.0	4.10	0.91	49.8	29.5	0.61	5.31	2.02	1.27	0.09	0.17	0.32
S26	75	20	6.35	0.45	543	137.0	57.1	9.50	4.95	11.2	8.29	0.85	58.6	23.4	0.49	14.6	0.69	1.10	0.07	0.19	0.25
S27	72	20	6.48	0.41	558	140.0	54.5	9.10	4.34	10.6	8.19	0.89	57.0	18.4	0.36	3.05	1.49	0.85	0.07	6.73	0.26
S28	76	21	5.91	0.83	471	172.0	139.0	46.5	08.88	54.7	1.89	0.66	27.0	12.5	0.41	6.32	2.03	1.60	0.09	4.00	4.59
S29	69	28	6.57	0.81	492	121.0	122.0	6.40	7.79	18.8	9.63	0.71	33.3	22.1	0.53	5.25	2.74	1.81	0.13	0.93	0.38
S30	86	16	4.61	0.57	467	196.0	158.0	7.70	10.40	19.8	4.90	0.99	74.2	22.5	0.19	2.16	1.13	0.67	0.07	0.28	0.20

Sam-	ple	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
S31	77	25	5.25	0.73	400	89.0	101.0	18.4	09.98	22.6	2.89	0.68	26.3	32.6	0.31	3.96	1.80	1.05	0.10	0.14	0.24	
S32	82	22	7.22	0.72	461	148.0	148.0	2.80	9.05	53.8	2.84	0.95	39.1	14.0	0.39	4.74	2.51	1.55	0.12	0.18	0.31	
S33	82	20	5.14	0.45	492	166.0	17.0	2.80	5.26	28.5	3.59	1.00	87.4	21.8	0.14	2.02	1.00	0.59	0.06	0.33	0.16	
S34	80	16	4.48	0.47	468	187.0	32.0	3.80	7.64	18.8	4.52	1.02	70.4	18.7	0.14	2.18	0.89	0.57	0.06	0.09	0.19	
S35	78	19	6.62	0.24	539	174.0	7.4	9.02	1.91	8.9	9.78	0.99	80.3	14.4	0.17	2.63	1.31	0.72	0.08	0.24	0.17	
S36	78	20	7.71	0.18	683	175.0	4.8	7.04	2.36	6.4	6.57	1.00	62.0	14.2	0.30	5.24	2.19	1.97	0.11	0.42	0.23	
S37	75	23	6.63	0.23	464	131.0	6.0	7.51	3.35	9.7	5.62	0.65	40.6	15.2	0.10	2.68	1.23	1.02	0.09	2.88	0.12	
S38	90	21	2.33	0.38	656	165.0	125.0	1.00	0.90	24.8	15.7	0.17	51.0	79.1	0.40	8.97	3.95	2.34	0.22	2.20	0.42	

Using R, summary statistics were created from table 1 (table 2) and initial observations were made. We found that concentrations for each element varied greatly in range and skew (if skew in this case is distance between mean and median) between each other. The elements Sb, As, Sn, Li, Sr, and Ba all show strong positive skews, numerous other elements show weaker skews. Cu and Zn are roughly symmetric in terms of skew. Ranges vary greatly with many magnitude differences between the ‘percentage based’ vs the ‘ppm’ elements (where percentage is $\sim 10^4$ more than ppm of the same value assuming the 2 measurement techniques are equally accurate) and between trace element ranges (Rb having a minimum of 0.03ppm while Ni has a maximum of 683ppm), and between the different samples vs same element (I.e. Sr has a minimum of 0.08ppm and a maximum of 15.10ppm which is over 180 times more than the minimum) these characteristics hint that a data transformation may be appropriate to increase interpretability.

Caponetti, Armetta, Brusca, et al. (2017) observed that samples S13, S25, and S38 were potential outliers due to various unusual element concentrations which we also saw. We also

found that S22, S18, and S17 appear to have high Sr content (15.1ppm, 11.8ppm, 7.27ppm respectively compared to a median of 0.42ppm) while S28, S22, and S19 have high Ba content (4.59ppm, 3.61ppm, 3.37ppm respectively compared to a median of 0.28ppm).

(MAYBE SAY 'further data clustering analysis may find that these groupings of samples cluster together in similar groups, furthering Caponetti, Armetta, Brusca, et al. (2017) claim that trace element analysis could be used to differentiate providence of alloy production)

Figure 2: Element Concentration Spread of 38 Orichalcum Ingots

38 Orichalcum Ingots | Notched Boxplots | Highlighted Outliers

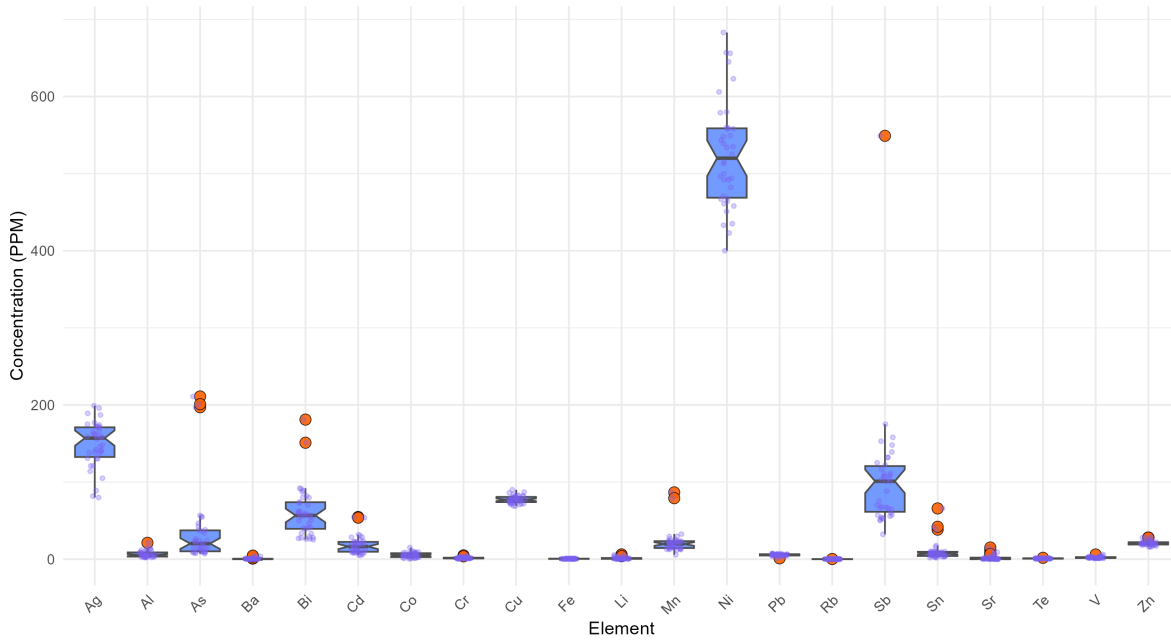


Figure 1: Notch Boxplot

Ni (nickel) had the widest interquartile range, indicating the greatest variation among the samples, while Sb (antimony) shows the largest total range, likely driven by sample 13's recorded Sb being magnitudes larger than any of the other samples.

Due to the fact the data was recorded at different scales (percentage and PPM) and obvious outliers, log transformation and scaling were the next logical steps.

3.1.1 Inconsistencies in Percentages

The percentages of elements Cu, Zn, Pb, and Fe do not add up to 100% for each sample, as shown in the summary table below.

Table 2: Summary statistics for main component elements.

Element	Mean	SD	Median	Min	Max
Cu	77.42	4.98	77.00	69.00	90.00
Zn	20.95	3.25	20.00	16.00	28.00
Pb	5.50	1.34	5.92	1.05	7.71
Fe	0.50	0.23	0.44	0.14	1.06
percentage_total	104.36	4.28	104.40	93.17	113.71

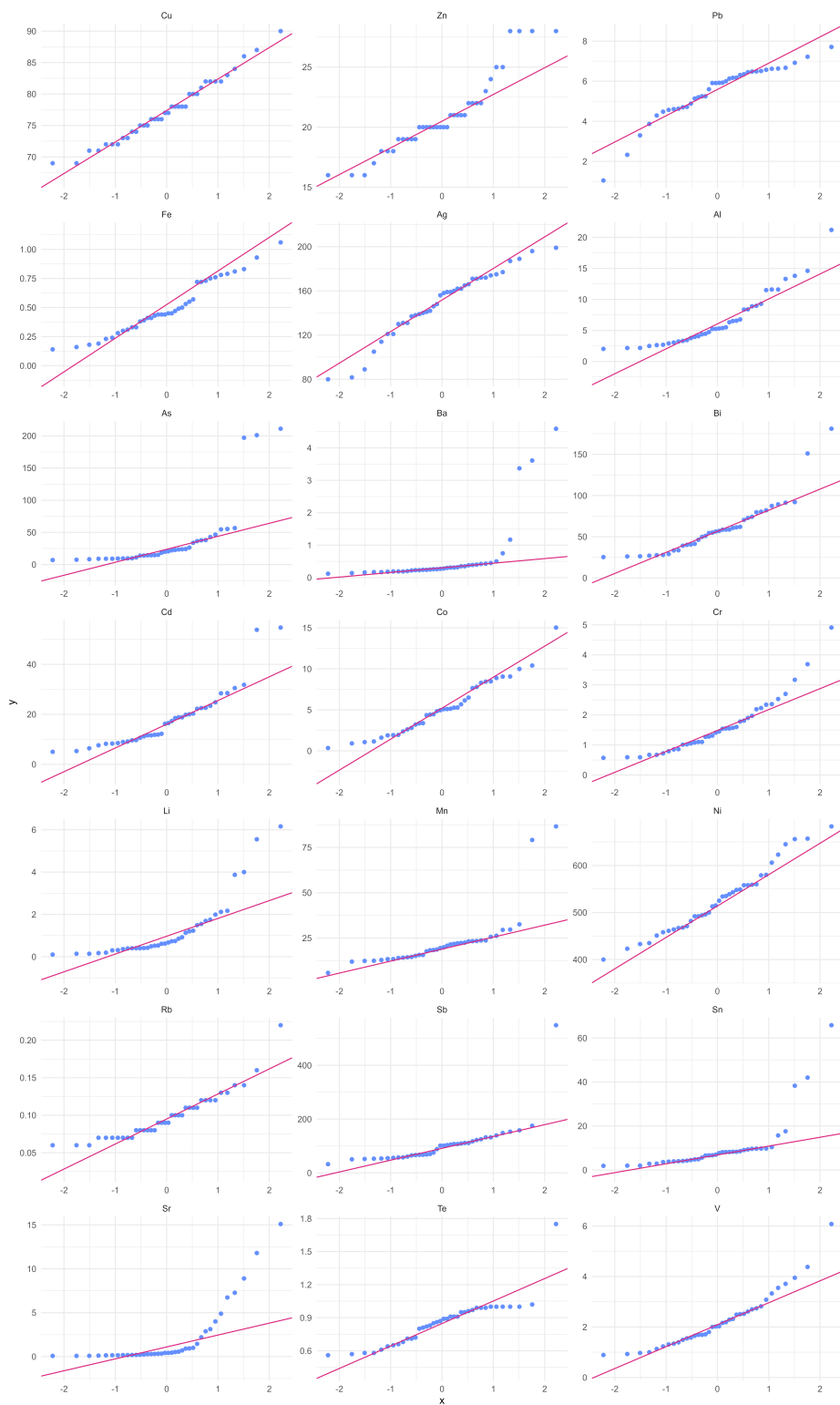
Trace elements weren't included for interpretability as the values are $\times 10^{-4}$ smaller.

These samples have a mean of 104.4\% with a range 93.17\% to 113.71\%. Where 33 of the 38 samples exceed 100\%. The most significant deviations are samples 8 and 38 (COMMENT ON THIS). Caponetti, Armetta, Brusca, et al. (2017) attribute this to be due to errors generated from the low quantity of available material for sampling for each observation.

These samples have a mean of 104.4\% with a range 93.17\% to 113.71\%. Where 33 of the 38 samples exceed 100\%. The most significant deviations are samples 8 and 38 which have percentage totals of 112.44\% and 113.71\% respectively. Caponetti, Armetta, Brusca, et al. (2017) attribute this to be due to errors generated from the low quantity of available material for sampling for each observation. For example, only 12.3 mg of sample 38 were obtained compared to the ideal 20mg of material as described in their methods.

3.2 QQ-plots & Henze-Zirkler Test For Multivariate Normality

We first compared the QQ-normal plots of table 1 and compared it to the log transformed table 1 to assess if log transformation would be appropriate to manage outliers and standardisation.



As this was before there was any scaling performed, we opted to display the elements measured in percentages and the elements measured in PPM separately to better display their individual influence.

Both Percentages and PPM show blatant departures from the assumption of normality. Few of the points follow the theoretical model line, steps/gaps in the points are likely because the various element concentrations have such different numerical ranges, both plots somewhat show an s-shaped curve which tend to indicate that the variables have heavy skew (extreme values/outliers) which is unsurprising considering the ranges and variations between elements being widely different. Overall significant departures from normality can be seen for both Percentages and PPM. We do note that the qq-plot for percentages is marginally better due to the more symmetric distribution of values.

```
#log transform data

logIngots <- log(as.matrix(Ingots_F_no[-1,-1]))
#print(logIngots)

knitr::kable(logIngots,
              caption = "Log Transformed Data",
              digits = 2,
              format = "pipe")
```

Table 3: Log Transformed Data

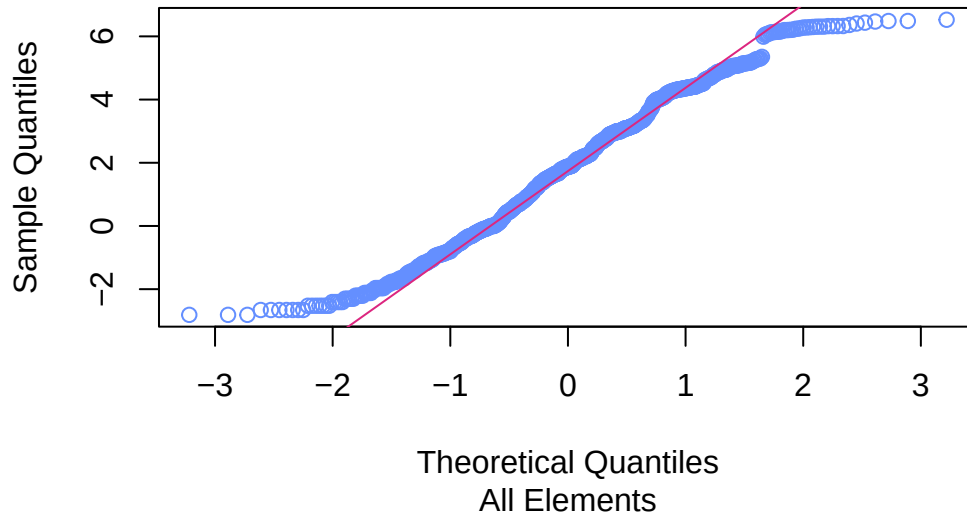
	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
2	4.47	2.77	1.46	-	6.28	5.29	4.88	3.59	2.13	3.00	1.57	0.00	4.29	3.03	0.77	1.50	0.44	0.27	-	-	-
				0.63															2.66	1.90	1.71
3	4.29	3.33	1.78	-	6.11	4.88	4.65	3.17	2.14	2.80	1.90	-	3.37	2.75	1.39	2.18	1.20	0.99	-	2.19	-
				0.25								0.45							1.97		1.05
4	4.38	3.00	1.82	-	6.32	5.14	4.25	2.41	1.57	2.48	2.23	-	4.11	3.14	1.82	2.62	1.81	1.59	-	-	-
				0.94								0.20							1.97	1.24	0.94
5	4.36	3.09	1.35	-	6.24	5.05	4.20	2.67	1.87	1.67	2.04	-	4.07	3.15	1.35	2.45	1.48	1.31	-	-	-
				0.60								0.54							2.12	0.09	0.80
6	4.39	3.09	1.87	-	6.36	5.07	4.10	2.12	0.63	2.26	2.14	-	4.12	2.75	0.75	1.47	0.83	0.64	-	-	-
				1.66								0.04							2.21	0.80	0.29
7	4.32	3.00	1.86	-	6.33	4.87	4.67	5.35	1.21	2.45	2.87	-	3.93	3.10	0.56	1.70	0.77	0.45	-	-	-
				1.11								0.22							2.04	1.08	0.69
8	4.41	3.18	1.81	-	6.31	5.08	4.32	2.27	1.16	2.03	1.94	-	4.07	3.17	0.69	2.59	1.27	0.86	-	0.37	-
				1.17								0.19							2.21		1.35
9	4.38	2.89	1.53	-	6.18	5.07	4.69	3.19	1.66	3.42	1.37	-	4.51	3.17	0.52	2.13	0.99	0.58	-	-	-
				0.82								0.09							2.66	2.21	0.84
10	4.28	3.33	1.78	-	6.08	4.65	4.71	3.16	2.12	2.79	1.89	-	3.32	2.97	0.21	1.91	0.69	0.34	-	-	-
				0.27								0.49							2.12	1.56	1.20
11	4.33	2.94	1.78	-	6.26	4.93	3.97	2.71	1.64	2.14	2.12	-	4.00	2.56	-	1.88	0.54	0.25	-	-	-
				0.84								0.15			0.08				2.30	0.58	1.43
12	4.23	2.83	1.90	-	6.49	4.95	3.91	2.62	1.49	2.12	2.10	-	4.01	2.60	0.18	1.17	0.50	0.10	-	1.14	-
				0.69								0.14							2.30		1.51

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
13	4.36	3.00	1.19	0.06	6.13	5.24	6.31	5.28	0.47	2.85	3.65	0.00	5.20	4.46	0.44	2.23	1.12	0.80	-	-	0.16
																			2.66	0.65	
14	4.30	3.22	1.55	-	6.07	4.40	4.48	2.68	1.63	3.01	0.68	-	3.26	3.39	0.12	1.87	0.91	0.44	-	-	-
				0.82							0.34								2.30	1.77	1.66
15	4.43	3.00	1.65	-	6.31	5.14	4.67	3.03	0.06	3.15	1.44	-	4.52	2.48	-	1.24	0.46	0.08	-	-	-
				1.97							0.05			0.30					2.53	2.53	1.77
16	4.41	2.94	1.55	-	6.33	5.09	4.63	2.99	0.13	3.10	1.39	-	4.41	2.52	-	3.05	0.84	0.93	-	-	-
				1.83							0.12			0.40					2.53	1.17	1.17
17	4.34	3.00	1.93	-	6.21	5.18	5.03	4.00	2.20	3.46	4.19	-	3.33	1.76	-	1.20	-	-	-	1.98	-
				0.24							0.21			0.92		0.07	0.40	2.53		1.05	
18	4.42	2.94	1.52	-	6.24	5.06	4.71	3.10	1.63	3.35	1.34	-	4.49	3.07	-	1.41	0.52	0.01	-	2.47	-
				0.82							0.01			0.65					2.41		1.31
19	4.26	3.33	1.66	-	6.20	4.74	4.62	3.64	1.81	2.92	2.34	-	3.51	2.87	0.40	2.12	0.96	0.43	-	1.59	1.21
				0.33							0.33								1.83		
20	4.33	2.94	1.84	-	6.28	4.93	3.94	2.64	1.48	2.10	2.09	-	4.03	2.60	-	1.07	0.29	-	-	-	-
				0.89							0.03			0.48				0.24	2.81	0.87	0.92
21	4.30	3.04	1.59	-	6.05	4.38	4.22	2.67	1.74	3.11	0.69	-	3.23	3.27	-	0.91	-	-	-	-	-
				0.71							0.09			0.87		0.03	0.53	2.66	0.36	1.47	
22	4.26	3.04	1.79	-	6.41	4.96	5.16	4.01	2.71	2.47	3.74	-	3.84	3.25	-	2.45	1.04	0.78	-	2.71	1.28
				0.07							0.58			0.16					2.12		
23	4.28	3.09	1.82	-	6.43	4.80	4.21	2.21	0.96	2.45	2.20	-	3.69	2.91	-	1.68	0.92	0.37	-	-	-
				1.11							0.54			0.31					2.21	2.53	1.97
24	4.33	2.89	1.87	-	6.47	5.15	3.96	2.24	0.66	2.21	2.28	0.00	4.38	2.68	-	1.32	0.33	-	-	-	-
				1.27										0.92				0.15	2.53	1.90	1.27

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
25	4.36	2.89	0.05	-	6.21	5.09	3.47	4.04	-	1.61	1.41	-	3.91	3.38	-	1.67	0.70	0.24	-	-	-
				0.29					1.08			0.09			0.49				2.41	1.77	1.14
26	4.32	3.00	1.85	-	6.30	4.92	4.04	2.25	1.60	2.42	2.12	-	4.07	3.15	-	2.68	0.52	0.10	-	-	-
				0.80								0.16			0.71				2.66	1.66	1.39
27	4.28	3.00	1.87	-	6.32	4.94	4.00	2.21	1.47	2.36	2.10	-	4.04	2.91	-	1.12	0.40	-	-	1.91	-
				0.89								0.12			1.02			0.16	2.66		1.35
28	4.33	3.04	1.78	-	6.15	5.15	4.93	3.84	2.18	4.00	0.64	-	3.30	2.53	-	1.84	0.71	0.47	-	1.39	1.52
				0.19								0.42			0.89				2.41		
29	4.23	3.33	1.88	-	6.20	4.80	4.80	3.27	2.05	2.93	2.26	-	3.51	3.10	-	1.66	1.01	0.59	-	-	-
				0.21								0.34			0.63				2.04	0.07	0.97
30	4.45	2.77	1.53	-	6.15	5.28	5.06	3.63	2.34	2.99	1.59	-	4.31	3.11	-	0.77	0.12	-	-	-	-
				0.56								0.01			1.66			0.40	2.66	1.27	1.61
31	4.34	3.22	1.66	-	5.99	4.49	4.62	2.91	2.30	3.12	1.06	-	3.27	3.48	-	1.38	0.59	0.05	-	-	-
				0.31								0.39			1.17				2.30	1.97	1.43
32	4.41	3.09	1.98	-	6.13	5.00	5.00	3.76	2.20	3.99	1.04	-	3.67	2.64	-	1.56	0.92	0.44	-	-	-
				0.33								0.05			0.94				2.12	1.71	1.17
33	4.41	3.00	1.64	-	6.20	5.11	4.76	3.13	1.66	3.35	1.28	0.00	4.47	3.08	-	0.70	0.00	-	-	-	-
				0.80											1.97			0.53	2.81	1.11	1.83
34	4.38	2.77	1.50	-	6.15	5.23	4.88	3.52	2.03	2.93	1.51	0.02	4.25	2.93	-	0.78	-	-	-	-	-
				0.76											1.97		0.12	0.56	2.81	2.41	1.66
35	4.36	2.94	1.89	-	6.29	5.16	4.05	2.20	0.65	2.19	2.28	-	4.39	2.67	-	0.97	0.27	-	-	-	-
				1.43								0.01			1.77			0.33	2.53	1.43	1.77
36	4.36	3.00	2.04	-	6.53	5.16	4.17	1.95	0.86	1.86	1.88	0.00	4.13	2.65	-	1.66	0.78	0.68	-	-	-
				1.71											1.20				2.21	0.87	1.47

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
37	4.32	3.14	1.89	-	6.14	4.88	4.19	2.02	1.21	2.27	1.73	-	3.70	2.72	-	0.99	0.21	0.02	-	1.06	-
				1.47								0.43			2.30				2.41	2.12	
38	4.50	3.04	0.85	-	6.49	5.11	4.83	5.30	-	3.21	2.75	0.56	5.02	4.37	-	2.19	1.37	0.85	-	0.79	-
				0.97					0.11						0.92				1.51	0.87	

Normal QQ-Plot for the log-Transformed Data



The QQ-plot of the log transformed data roughly followed the theoretical line (in red), this is evidence that the normality assumption may hold. The tails are relatively long and flat, indicating that there may be some extreme outliers. There are several steps in the upper and lower tails, which alone is not necessarily a departure from normality, and may be explained by the zero values returned for some Sample's variables after the log transformation.

The observed zero-values for (PUT SAMPLES NUMBERS IN HERE) in (TABLE) in variables(HERE), which while not a large quantity of the samples, still provided some concern

as $\log(0)$ is undefined and can influence non-normality in multivariate data. The negative values are not an issue because we will be using metrics which are calculated from coordinate distances for clustering, and can handle negative values.

We chose to combat the zero values by standardising the data set using $\log(x + 1)$, where x is our data. This, coupled with a Henze-Zirkler test, established if there is preliminary evidence for multivariate normality.

Table 4: Log Transformed Data

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba	
2	4.472	3.332	6.381	8.320	8.722	7.833	7.033	6.926	7.449	6.645	5.237	8.583	1.372	3.559	1.915	3.769	6.449	6.083	7.206	7.658	6.651	144
3	4.304	3.667	2.962	9.576	6.133	6.882	8.602	3.429	2.448	1.862	2.041	2.249	4.962	1.873	4.609	2.379	5.245	5.678	3.338	2.935	3.401	1046
4	4.394	4.044	5.209	6.373	6.351	4.974	4.942	6.893	2.763	2.157	2.237	6.598	3.032	8.863	5.685	6.046	9.272	7.706	4.580	2.836	3.223	3037
5	4.363	4.185	4.598	3.093	8.234	2.236	2.462	5.748	2.840	3.568	5.562	1.737	4.088	3.320	4.578	3.093	6.987	2.688	4.132	6.808	7.103	25636
6	4.406	7.192	4.016	2.353	3.363	3.028	5.174	5.782	2.163	7.790	2.835	4.072	2.043	5.855	4.097	8.678	3.969	0.866	7.107	3.670	5.539	6158
7	4.336	7.034	5.208	2.286	1.789	9.873	1.087	3.135	6.536	2.763	9.973	1.587	7.967	2.148	4.631	6.879	8.050	5.720	0.392	2.706	6.496	4651
8	4.418	3.218	8.764	3.270	0.308	5.098	1.483	3.986	6.498	2.706	7.626	9.684	3.066	9.766	8.095	2.664	2.595	1.271	1.400	3.890	0.880	1117
9	4.394	4.944	4.398	1.694	6.480	0.751	7.043	8.268	8.173	7.009	8.833	6.071	6.320	2.468	9.895	2.107	3.083	0.281	6.076	5.863	6.607	6744
10	4.296	4.367	2.296	0.155	5.303	3.660	3.479	3.199	8.672	8.840	9.094	7.066	2.343	4.070	4.862	0.056	2.098	6.879	6.263	3.290	6.262	3643
11	4.343	3.895	7.933	5.597	6.764	3.393	4.794	2.687	8.831	3.219	5.222	8.962	0.576	6.363	0.865	2.303	2.199	6.988	0.109	3.310	2.685	1114
12	4.248	4.890	3.073	7.406	6.468	2.295	5.827	3.769	4.607	5.230	2.126	6.625	9.082	3.607	2.288	4.439	8.345	5.900	9.953	4.082	7.985	509
13	4.363	4.044	5.238	6.730	7.000	0.247	0.349	9.288	2.955	9.069	6.712	6.531	2.020	4.727	8.360	9.342	0.060	9.701	4.687	6.886	7.703	7272
14	4.317	4.288	0.970	1.662	6.033	0.415	2.202	8.747	2.805	0.048	7.097	0.236	1.993	5.324	0.763	0.210	2.236	9.903	7.009	3.316	7.003	39533
15	4.442	6.644	5.824	5.493	3.389	0.187	4.082	1.072	6.722	7.004	5.636	3.007	8.392	5.594	0.153	8.890	6.947	7.892	3.670	6.700	6.570	0037
16	4.418	3.995	7.732	3.968	8.130	6.709	3.764	3.393	9.746	0.805	1.162	9.468	6.570	6.045	5.528	2.060	9.999	0.682	9.700	0.700	2.700	271
17	4.356	7.004	5.209	3.988	2.221	5.606	5.784	6.993	2.980	8.567	4.290	1.759	3.336	8.892	6.936	4.700	9.687	5.300	8.036	9.610	6.340	1046
18	4.430	2.977	7.732	7.964	6.236	1.068	9.043	1.999	8.866	3.809	9.586	6.681	3.061	3.835	4.587	6.033	9.895	6.981	0.862	5.791	2.390	169

	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba
19	4.2766367286258133203578496249766119657978443365132334389248521228374283707381648427024607630																				
20	4.3438095173292333588711941364608768068822922099367380838366722824263585015892056268063604722																				
21	4.31748811072153087069739442921074389731957009526373106253968370636990728036370667658062870142																				
22	4.2766001104230607520852028170448306972162047612004685852846645218366970250002093278885282279																				
23	4.2904595149712298517801804022243545283767386979565731718008015481253112589169008808607096300283																				
24	4.3438054240935086860708063292263246878210925354960313702739723645002831203050765963070408601																				
25	4.3624484043978398620850937502565522972679076902044710328967727628421136528697096175700276317																				
26	4.33070345224707150909927264218511375339024238969518506504539877672989574110067658692331436																				
27	4.290469452222328589619987602333256752257202581636506649652707489879722627018606608610891117																				
28	4.343809110432969063166079329216802720050299801257683322662690359906108502650861607743299793																				
29	4.2484962203419933208589408121812547369826363658061933113983352677532908361823267653000835																				
30	4.46298832734546707464083204806358483603497349688132615170007395305726132208066638686823216																				
31	4.3567098087258451293969986249765223960764245845987906887452600071402967978098310702831114																				
32	4.41881352490655023233566390639769630767066905166738943768030930737429569600933328552740271																				
33	4.4188044582483775006509797706869685418823523860314828726763110283269314637038208917884200																				
34	4.392489327010812630623648903349261562108668083783096829806191102808636576810756208017739533																				
35	4.3624498270307263512945664786723645868239253776988139621734365770080233724733076901011570037																				
36	4.363408452040176855527058048866084429120004804019331472173212962363898600088560086063070142																				
37	4.330733806420808014208828004693112420235828900960377321085009533029862070600986135783353287																				
38	4.510869110202932308856841198362802631833037454097600983883273621799559938069708856913686569																				

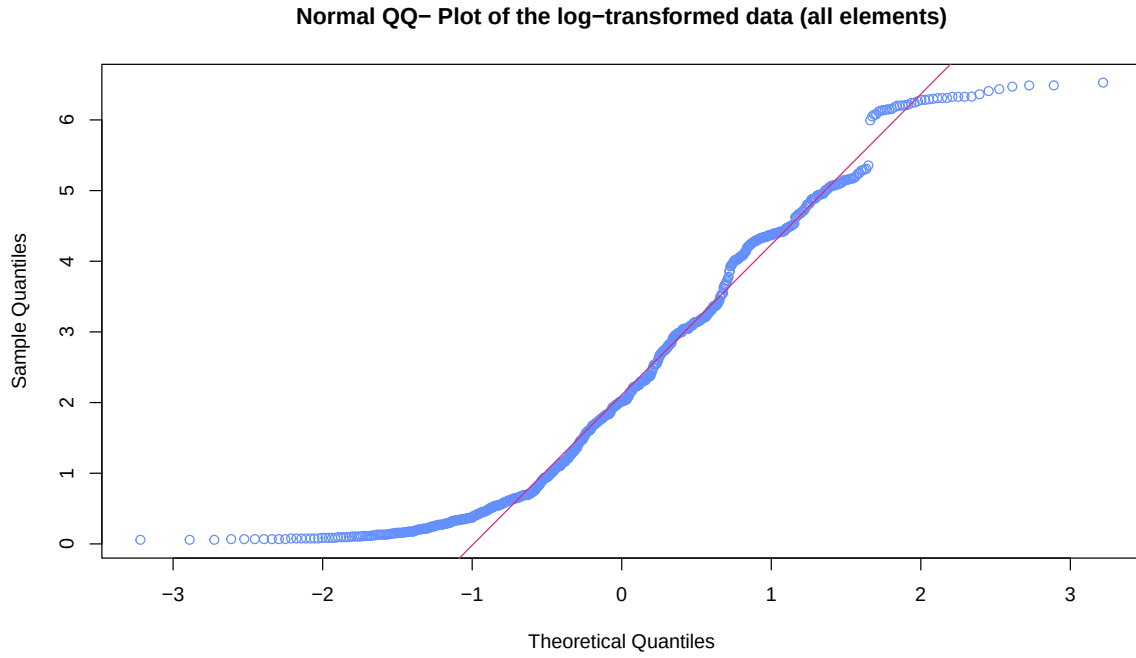


Table 5: Henze-Zirkler Test For Multivariate Normality

Test	Statistic	p.value	Method	MVN
Henze-Zirkler	1	0.069	asymptotic	Normal

As you can see in table 2, our values are now all non-zero.

The normal QQ-plot looks less normal after standardisation, with the length of the tails being accentuated and continuing presence of stepped data.

The Henze-Zirkler test stipulates

$\mathcal{H}_0$: The data follow a multivariate normal distribution,

$\mathcal{H}_1$: The data do not follow a multivariate normal distribution.

This generated a p-value=0.069, suggesting that there is insufficient evidence to reject that the data follow a multivariate normal distribution. However, the QQ-plot shows enough <flat tails, step, steep, etc> that further testing is warranted before determining if multivariate normality is likely.

Given the uncertainty of multivariate normality and the presence of outliers, further data manipulation by Mediod Absolute Deviation was determined to be the next best step towards multivariate analysis.

3.3 MAD

MAD was used because it is a robust, outlier-resistant, measure of variability. The log transformation somewhat fixed the skew and pulled the outliers in, but did not eliminate their mathematical influence.

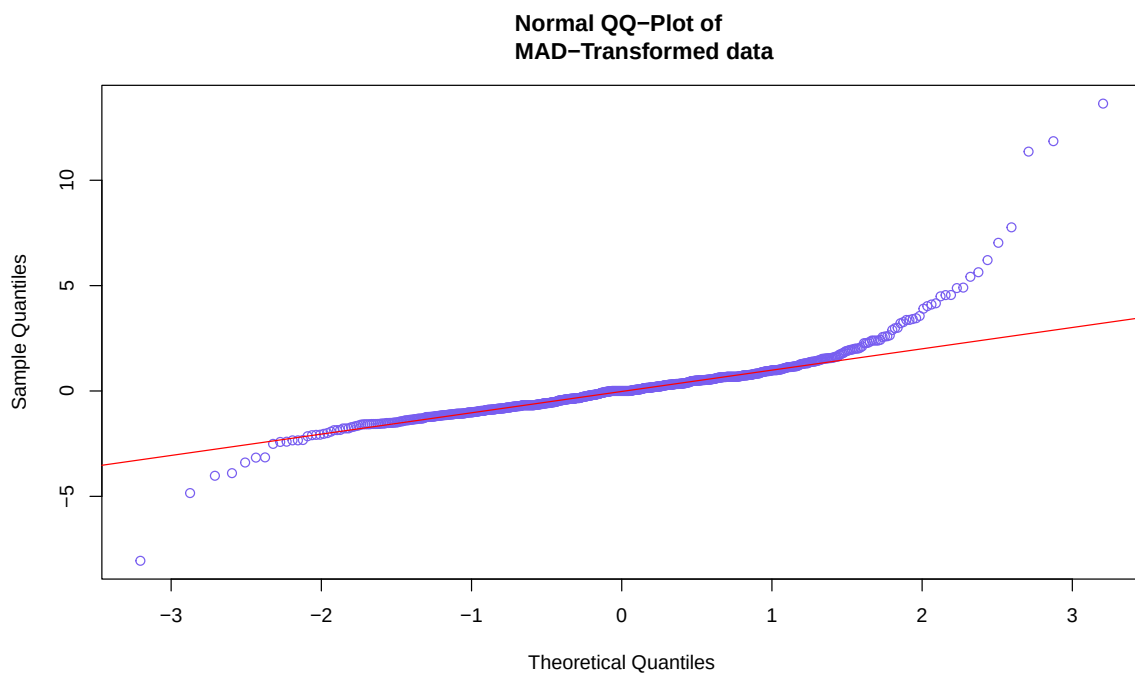
Table 6: Median / MAD

	Median	Mad	Median_over_Mad
Zn	3.04	0.13	22.57
Pb	1.93	0.15	12.81
Fe	0.37	0.15	2.47
Ni	6.25	0.12	50.39
Ag	5.07	0.17	30.24
Sb	4.62	0.56	8.23
As	3.07	0.86	3.55
Co	1.81	0.63	2.87
Cd	2.86	0.58	4.97
Sn	2.08	0.47	4.39
Te	0.64	0.08	8.33

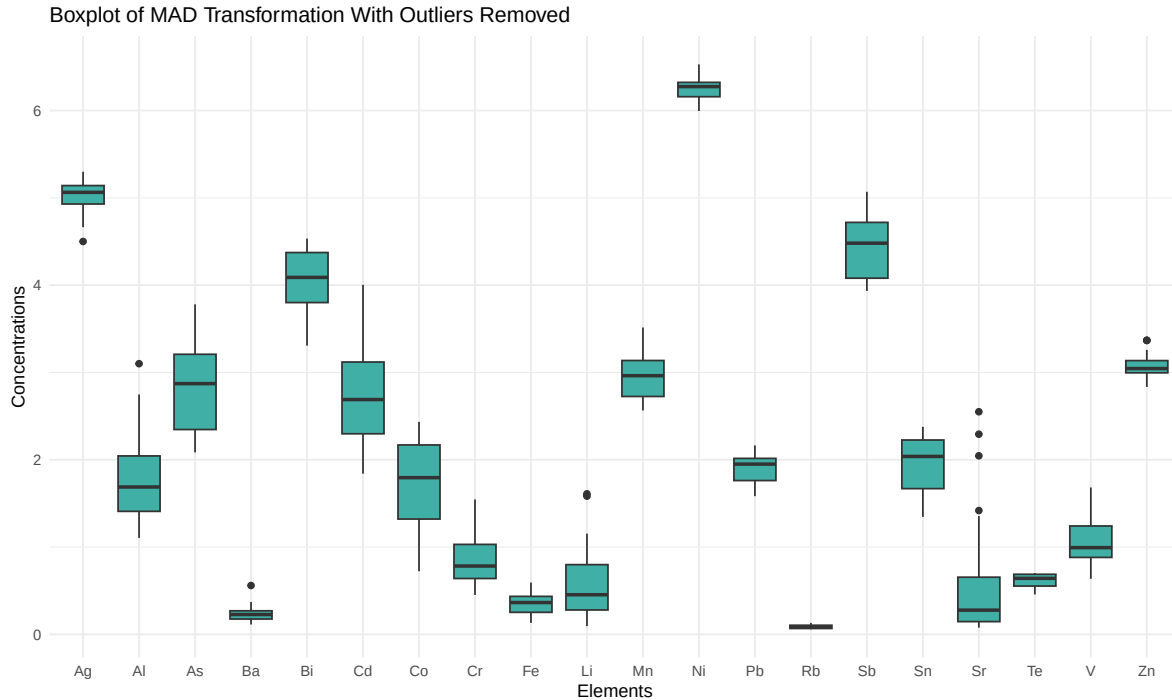
	Median	Mad	Median_over_Mad
Bi	4.06	0.50	8.11
Mn	3.02	0.35	8.65
Li	0.48	0.33	1.48
Al	1.83	0.60	3.06
V	1.11	0.29	3.86
Cr	0.88	0.27	3.21
Rb	0.09	0.03	3.14
Sr	0.35	0.31	1.12
Ba	0.25	0.11	2.28

From {TABLE NUMBER} we observed some massive deviations, as expected.

Next the robust Z score was used to identify outliers that were candidates for removal. For our cutoff, we went with Leys et al. (2013) recommendation, who argued that the MAD is the most robust scale measure under contamination and recommend a median plus or minus 2.5 time MAD rule for outlier detection.



Cu	Zn	Pb	Fe	Ni	Ag	Sb	As
4.3694479	2.9444390	0.7178398	0.5596158	6.2085900	5.0937502	3.5025499	4.0552572
Co	Cd	Sn	Te	Bi	Mn	Li	Al
0.2926696	1.7917595	1.6292405	0.6471032	3.9278964	3.4177267	0.4762342	1.8421357
V	Cr	Rb	Sr	Ba			
1.1052568	0.8197798	0.0861777	0.1570037	0.2776317			



The QQ plot shows very skewed data which we would expected...

What is after we have “cleaned” up the outliers our $|z| > 3$ rule removes samples 13, 25, and 38 which is exactly the sample ingots that Caponetti et al. set aside as their G4 group.

{should probably explain why cutting out 11 samples will be good for the rest of our analysis, something along the line of outliers, terrible samples and low sample quality}.

2.4 Multivariate normal distribution

What tests did we use? Henze Zirkler

Why did we use those tests?

Henze_Zirkler Test Results:

	Test	Statistic	p.value	Method	MVN
1	Henze-Zirkler	1.001	<0.001	asymptotic	Not normal

Henze_Zirkler Test Results:

	Test	Statistic	p.value	Method	MVN
1	Henze-Zirkler	1	0.055	asymptotic	Normal

Due to our Henze Zirkler test for both the un-transformed data (p-value <0.001) and the transformed data (p-value 0.046) we can safely reject the null hypotheses and conclude that the data regardless if it is transformed or not does not follow a multivariate normal distribution.

As our data isn't multivariate it isn't appropriate to perform a mahalanobis distance. {could potentially do need to research tho}

2.5 Correlation plots before and after

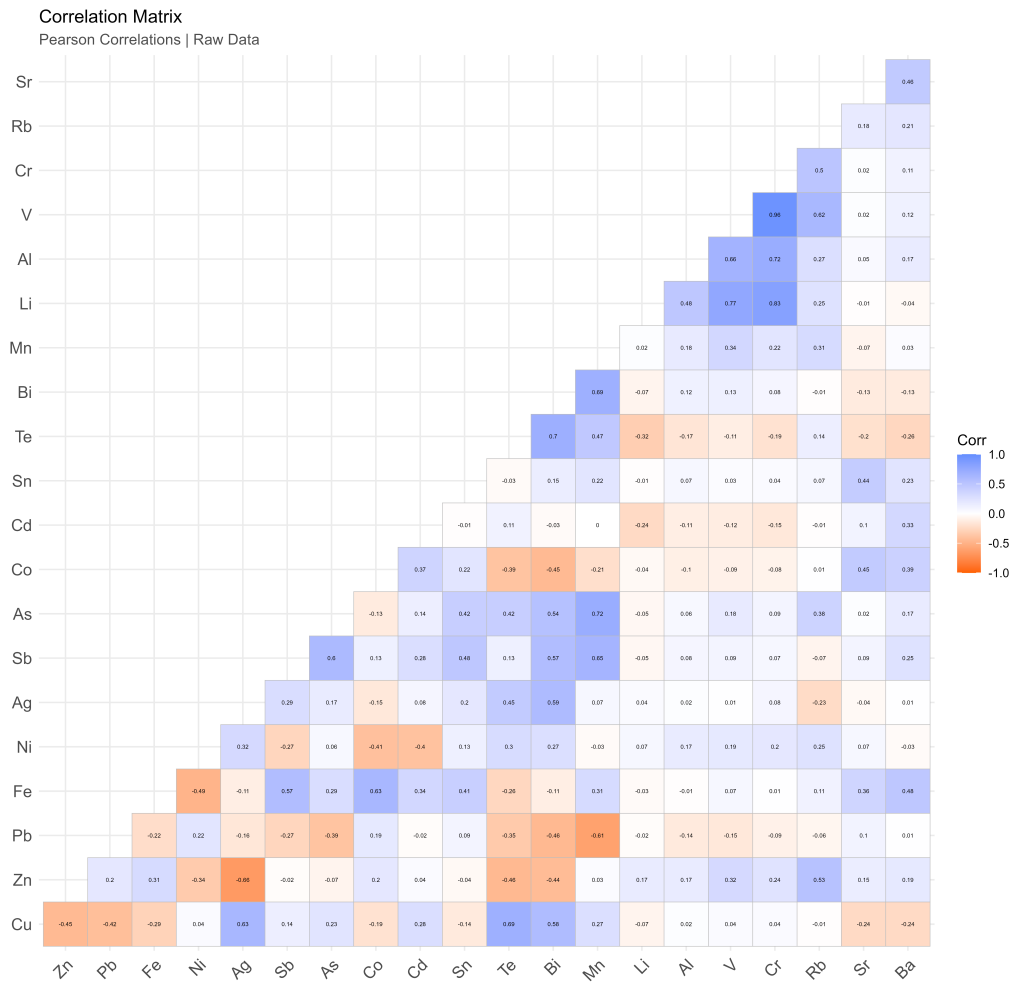


Figure 2: Correlation matrix of Raw Data

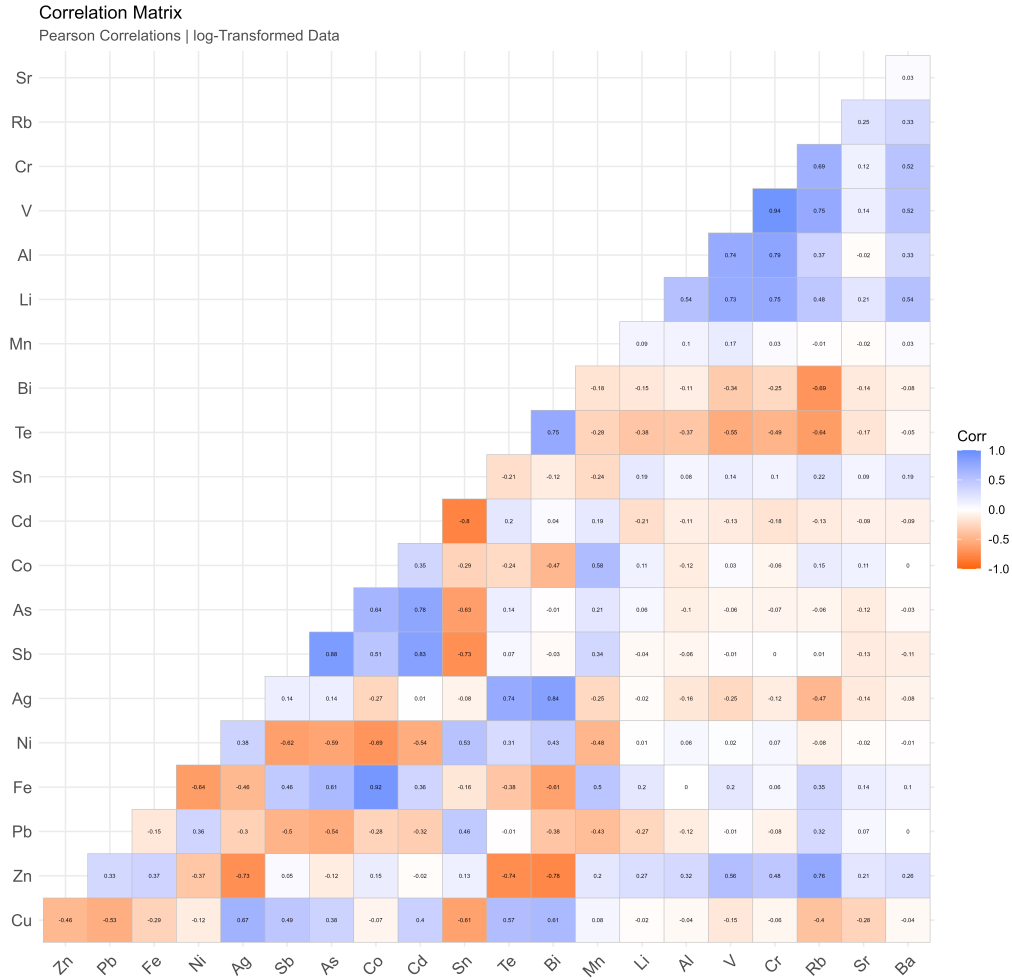


Figure 3: Correlation matrix of log-Transformed Data

Seeing a lot more correlation in the data with the massive outliers removed. Correlation plot some things to highlight; is that the strong positive association between V and Cr is also strongly positively association in the data with the outlying samples removes.. The transformed data also clearly shows a strong positive association for Bi and Ag which the un-transformed data doesn't show as significantly. the negative association between Zn and Te becomes more stronger in the transformed data.

The transformation didn't just strength some of the associations it also happened to reverse the

sign for several of the relationships, Bi and Mn was one the raw data had a positive association but the transformed data turned it into a negative association similar with the relation ships between Sb and Sn as well as Rb and Te. It went the other direction (negative to positive) for As and Co. This is a positive sign that the robustness of the transformation, i.e removing some of the outliers demonstrates their influence on the raw data.

Given the uncertainty about multivariate normality, a linear data analysis was chosen over a primary component analysis.

4. LDA

4.1 PAM

We started by performing a PAM (Partitioning Around Medoids) analysis

Table 7: PAM clustering summary across candidate values of k.

k	Small-			Cluster_Sizes
	Avg_Sil- houette	est_Clus- ter	Largest_Clus- ter	
2	0.182	7	19	7, 19
3	0.246	5	15	6, 5, 15
4	0.162	5	10	6, 5, 10, 5
5	0.179	4	7	6, 5, 4, 7, 4
6	0.176	3	7	5, 5, 3, 3, 7, 3
7	0.164	1	7	5, 4, 3, 3, 7, 3, 1
8	0.170	1	7	5, 4, 2, 1, 3, 7, 3, 1
9	0.168	1	7	4, 4, 2, 1, 2, 7, 2, 3, 1
10	0.152	1	7	4, 4, 2, 1, 1, 7, 2, 1, 3, 1

Small-				
	Avg_Sil-	est_Clus-	Largest_Clus-	
k	houette	ter	ter	Cluster_Sizes
11	0.147	1	7	4, 1, 2, 1, 1, 3, 7, 2, 1, 3, 1
12	0.128	1	7	4, 1, 1, 1, 1, 1, 3, 7, 2, 1, 3, 1
13	0.123	1	6	4, 1, 1, 1, 1, 1, 3, 6, 2, 1, 3, 1, 1
14	0.113	1	6	4, 1, 1, 1, 1, 1, 3, 6, 1, 1, 1, 3, 1, 1
15	0.116	1	6	4, 1, 1, 1, 1, 1, 2, 6, 1, 1, 1, 3, 1, 1, 1
16	0.126	1	5	4, 1, 1, 1, 1, 1, 2, 5, 1, 1, 1, 1, 3, 1, 1, 1
17	0.100	1	4	4, 1, 1, 1, 1, 1, 2, 4, 1, 1, 1, 1, 3, 1, 1, 1, 1
18	0.095	1	4	1, 1, 1, 1, 1, 1, 2, 4, 1, 1, 1, 1, 3, 1, 3, 1, 1, 1
19	0.103	1	4	1, 1, 1, 1, 1, 1, 2, 4, 1, 1, 1, 1, 2, 1, 3, 1, 1, 1, 1
20	0.110	1	3	1, 1, 1, 1, 1, 1, 2, 3, 1, 1, 1, 1, 2, 1, 1, 3, 1, 1, 1, 1
21	0.107	1	3	1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 2, 1, 1, 3, 1, 1, 1, 1
22	0.099	1	2	1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 2, 1, 1, 2, 1, 1, 1, 1, 1
23	0.093	1	2	1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 2, 1, 1, 1, 1, 1
24	0.066	1	2	1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1
25	0.035	1	2	1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1

```

cluster size ave.sil.width
1      1      8      0.28

```

2	2	5	0.17
3	3	13	0.22

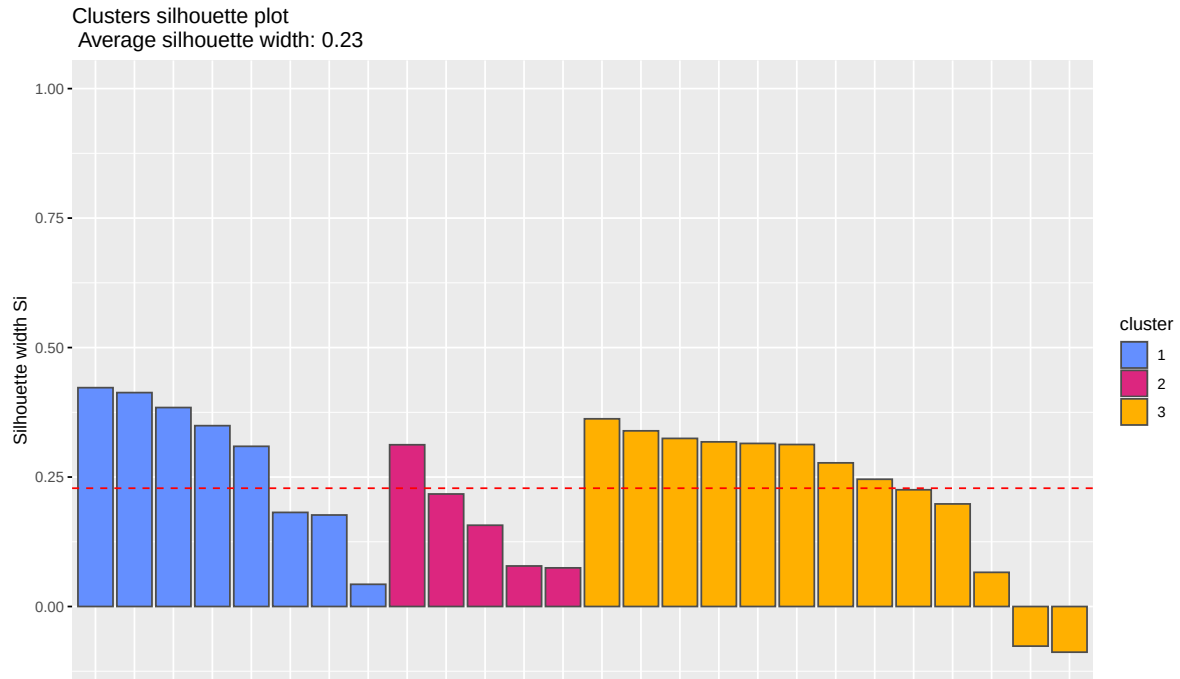


Table 8: Top ten cluster choices ranked by average silhouette width.

k	Avg_Silhou-	Largest_Clus-		
	ette	Smallest_Cluster	ter	Cluster_Sizes
3	0.246	5	15	6, 5, 15
2	0.182	7	19	7, 19
5	0.179	4	7	6, 5, 4, 7, 4
6	0.176	3	7	5, 5, 3, 3, 7, 3
8	0.170	1	7	5, 4, 2, 1, 3, 7, 3, 1
9	0.168	1	7	4, 4, 2, 1, 2, 7, 2, 3, 1
7	0.164	1	7	5, 4, 3, 3, 7, 3, 1

k	Avg_Silhou-	Largest_Clus-		
	ette	Smallest_Cluster	ter	Cluster_Sizes
4	0.162	5	10	6, 5, 10, 5
10	0.152	1	7	4, 4, 2, 1, 1, 7, 2, 1, 3, 1
11	0.147	1	7	4, 1, 2, 1, 1, 3, 7, 2, 1, 3, 1

The silhouette value/metric is a measurement of the quality of a clustering. between -1 and 1. The data reports the optimal silhouette width as 5, with 0.6496032, indicating some. There are several other clustering values with a reported similar quality.

Due to the ambiguous clustering, we opted to perform more cluster analyses to determine if this was a good fit.

MANHATTAN NOT EUCLEDIAN.

3.3 Cluster Analysis

Hierarchical cluster using Manhattan Distance as a measure. Because more robust to outliers then Euclidian etc.

PAM Clustering Summary

Average silhouette width and cluster sizes

k	Avg_Silhouette	Smallest_Cluster	Largest_Cluster	
2	0.182	7	19	
3	0.246	5	15	
4	0.162	5	10	
5	0.179	4	7	
6	0.176	3	7	
7	0.164	1	7	
8	0.170	1	7	
9	0.168	1	7	4,
10	0.152	1	7	4, 4,
11	0.147	1	7	4, 1, 2,
12	0.128	1	7	4, 1, 1, 1,
13	0.123	1	6	4, 1, 1, 1, 1,
14	0.113	1	6	4, 1, 1, 1, 1, 1,
15	0.116	1	6	4, 1, 1, 1, 1, 1, 2,
16	0.126	1	5	4, 1, 1, 1, 1, 1, 2, 5,
17	0.100	1	4	4, 1, 1, 1, 1, 1, 2, 4, 1,
18	0.095	1	4	1, 1, 1, 1, 1, 1, 2, 4, 1, 1,
19	0.103	1	4	1, 1, 1, 1, 1, 1, 2, 4, 1, 1, 1,
20	0.110	1	3	1, 1, 1, 1, 1, 1, 2, 3, 1, 1, 1, 1,
21	0.107	1	3	1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1,
22	0.099	1	2	1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 2,
23	0.093	1	2	1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2,
24	0.066	1	2	1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1,
25	0.035	1	2	1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,

Hierarchichal Cluster Dendrogram using Manhattan Distance

Average Linkage | Manhattan Distance | CCC = 0.798

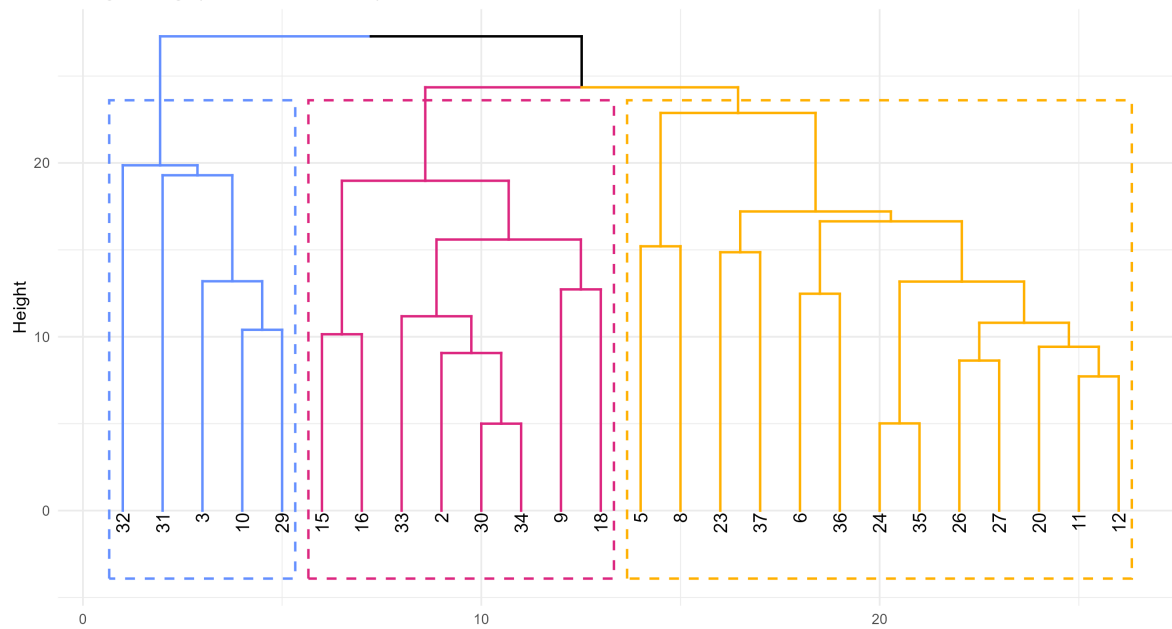


Figure 4: Dendrogram

```
d <- dist(scaled, method = "manhattan")
```

```
HiMan <- hclust(d, method = "average")
```

```
rownames(scaled) <- rownames(ICN)
```

```
p <- fviz_dend(
  HiMan,
  k = 5,
  rect = TRUE,
  k_colors = my_col,
  rect_border = my_col
)
```

```
)
```

```
head(ggplot_build(p)$data[[2]]$label)
```

```
[1] 32 31 3 10 29 15
```

```
26 Levels: 32 31 3 10 29 15 16 33 2 30 34 9 18 5 8 23 37 6 36 24 35 26 ... 12
```

```
length(unique(ggplot_build(p)$data[[2]]$label))
```

```
[1] 26
```

Something interesting top note is that for both cluster options they both cluster sample's 15, 10, and 22 this could be because of ...

Cophenetic correlation coefficient 0.825 is pretty dang good. I'm going to remake this using ggplot, should be nicer.

3.4 Fuzzy Analysis

As the third and final method of clustering, we performed a Fuzzy analysis.

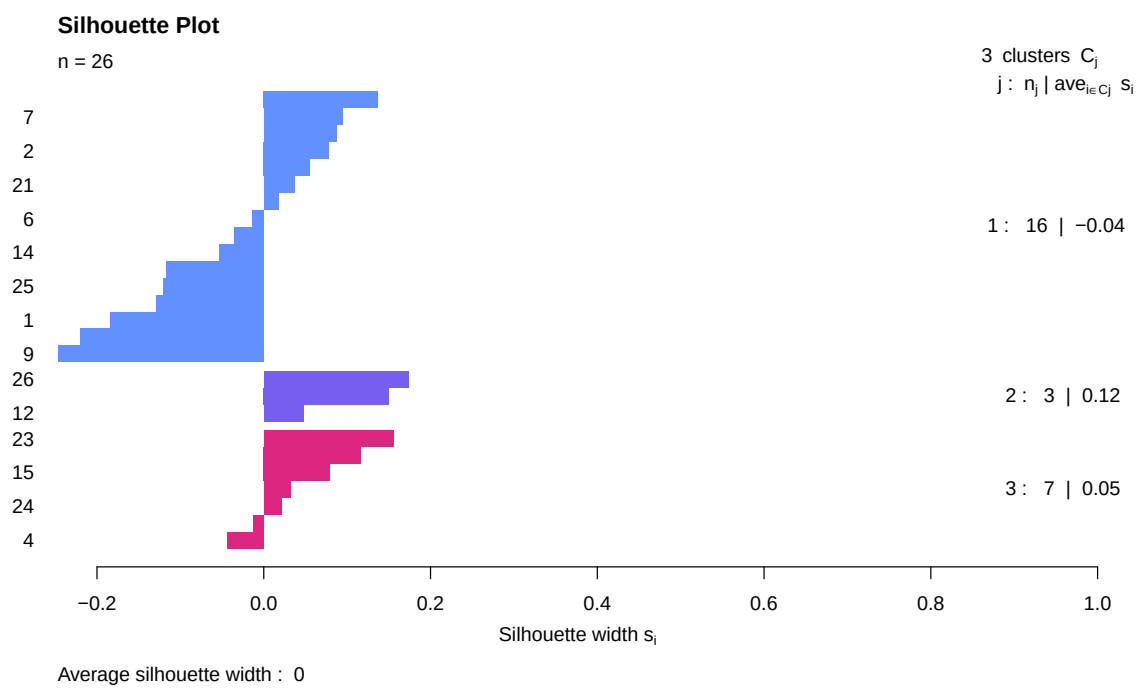
Table 9: Fuzzy K Silhouettes Clusterings

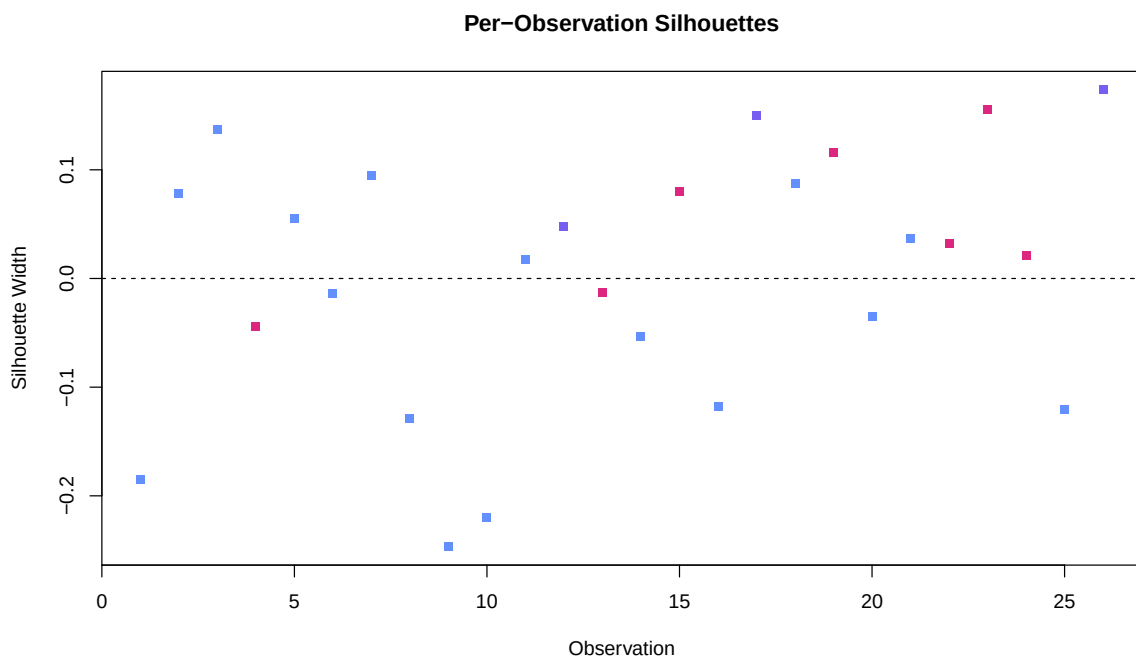
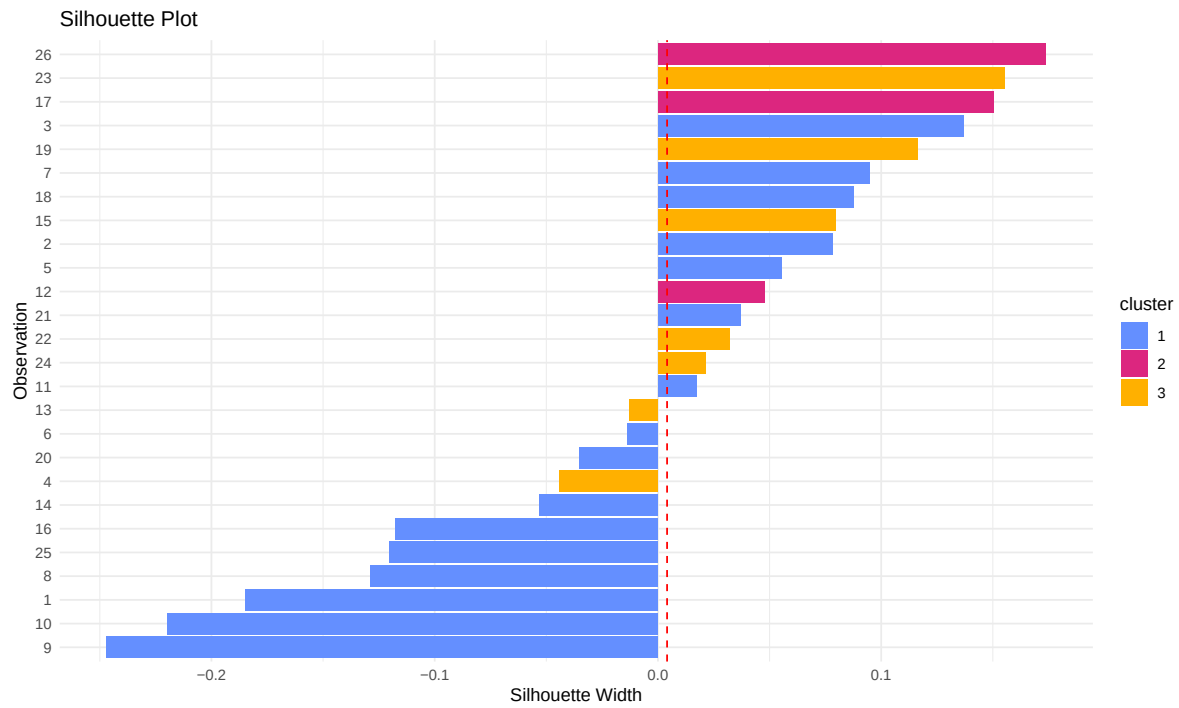
	Clus 1	Clus 2	Clus 3
2	0.3830127	0.2855148	0.3314725
3	0.4172427	0.3160363	0.2667210
5	0.4369832	0.2833827	0.2796341
6	0.3556217	0.2718466	0.3725317

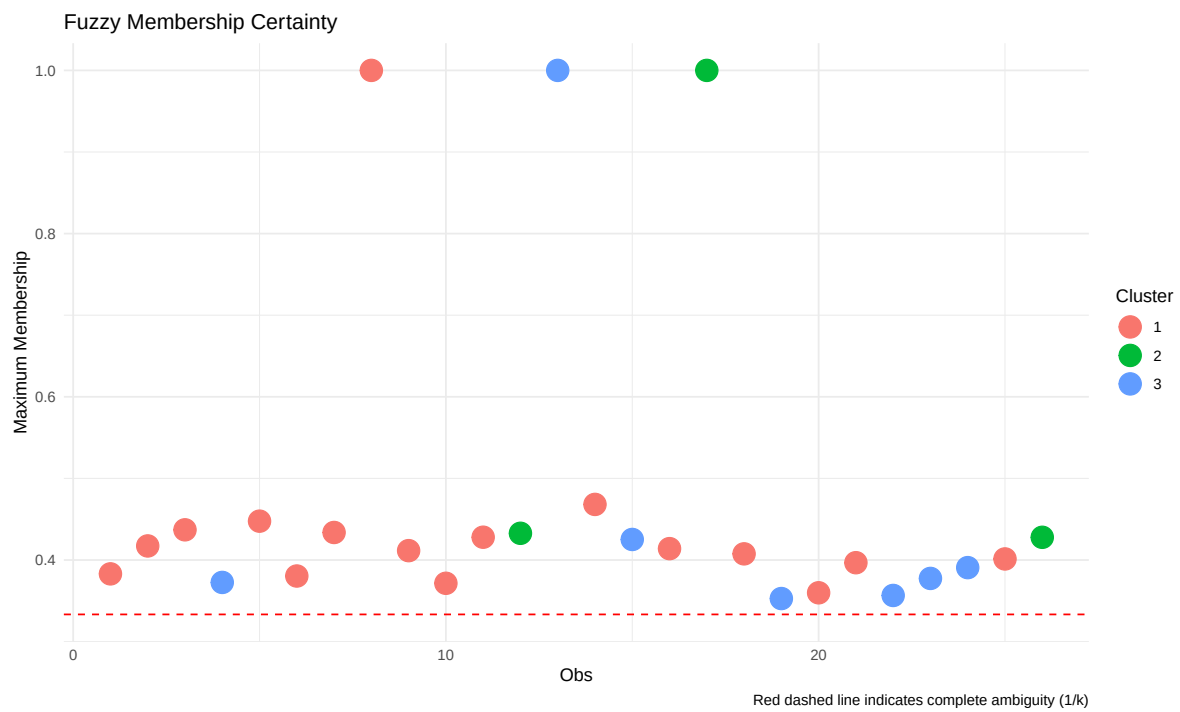
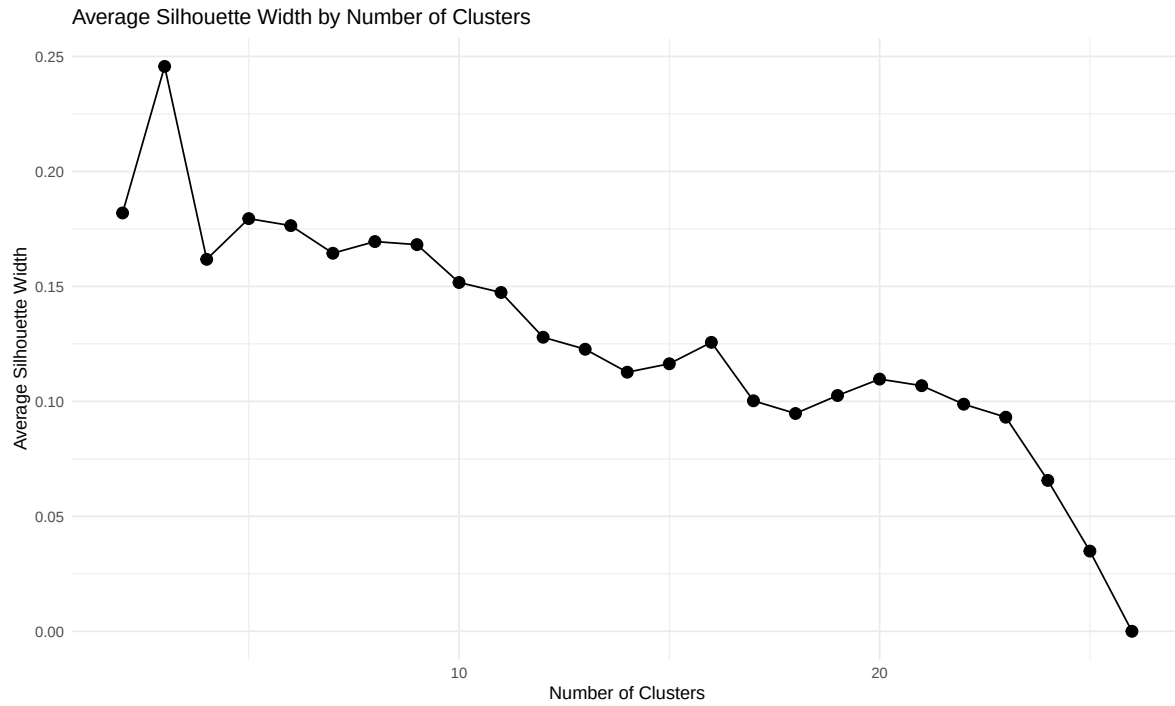
	Clus 1	Clus 2	Clus 3
8	0.4476196	0.3071149	0.2452655
9	0.3803899	0.2864301	0.3331800
10	0.4337303	0.2811057	0.2851640
11	1.0000000	0.0000000	0.0000000
12	0.4114799	0.3356098	0.2529103
15	0.3714748	0.2809841	0.3475411
16	0.4279359	0.2666536	0.3054105
18	0.2980514	0.4327364	0.2692122
20	0.0000000	0.0000000	1.0000000
23	0.4681284	0.2897138	0.2421578
24	0.3019973	0.2728233	0.4251794
26	0.4139329	0.2995528	0.2865142
27	0.0000000	1.0000000	0.0000000
29	0.4075424	0.3100154	0.2824422
30	0.3299705	0.3172533	0.3527762
31	0.3598310	0.3322387	0.3079303
32	0.3967004	0.2871549	0.3161447
33	0.2964350	0.3470044	0.3565606
34	0.3204017	0.3021561	0.3774423
35	0.3025122	0.3068187	0.3906691
36	0.4013054	0.3023600	0.2963346
37	0.3197456	0.4278293	0.2524251

The default value alpha=1 has been set

	cluster	neighbor	sil_width
[1,]	1	3	-0.18459765
[2,]	1	2	0.07818199
[3,]	1	2	0.13681298
[4,]	3	1	-0.04432890
[5,]	1	2	0.05528040
[6,]	1	3	-0.01374792







The graph of Fuzzy Membership Certainty shows that the majority of our observations are

near ambiguous in there assigned groupings.

Picks by median centrality, determines closest group and percentage it belongs in said group.

5. Results

We are unable to make direct

Caponetti, Armetta, Brusca, et al. (2017) determined that the quantity of sample was attributed to error generation. Both samples S38 and S08 had respective quantities 12.3mg and 16.0mg. The average amount of material sampled was reported at ~20mg.

The original study only had access to 38/40 ingots as they were on loan to another institute. This is an extension of that analysis, increasing the sample size at this point of the data analysis would inflate the false positive rate.

6. Discussion

Caponetti, Armetta, Martino, et al. (2017) determined 3 distinct groupings of ingots, which provides evidence of a scarce and highly prized alloy being manufactured in one place at one time. This was determined with the added variable of the physical form of the ingots.

Our results

The initial analysis was performed by non-invasive XRF, a method of getting accurate readings within a certain range of elements (find that bit again). The ICP-OE and ICP-MS chemometric techniques require a small amount of material to be sacrificed in order to obtain a full range of the elemental composition of each ingot.

The find was dated to ~600BCE, the approximate age of the ship, and by the pottery also found in the wreck.

A study on a sample group this large has never been done before. Also, this dataset is not very large. - love this sentence 10/10.

XRF is a portable, non-invasive method of getting accurate readings within a certain range of elements (find that bit again). The ICP-OE and ICP-MS chemometric techniques require a small amount of material to be sacrificed in order to obtain a full range of the elemental composition of each ingot.

Caponetti, Armetta, Brusca, et al. (2017) performed chemoanalysis to determine the trace elements of these orichalcum ingots. Using this data, they determined correlation between the observed different features and the composition to group the ingots into their appropriate cluster. This is an extension from a fledgling study by the same author, where they performed a less invasive XRF (X-Ray Fluorescence) spectroscopy of the ingots, a non-destructive test that can be performed in the field.

Their motivation was to create a starting point for giving insight into archaeometric study of the orichalcum ingots regarding their origin and manufacture technologies.

Our study also has the same motivations. We aim to provide a high level data analysis to provide a detailed, and logical, path to a linkage analysis.

Motivation, problem statement, approach, results, conclusions.

What were the limitations of this study?

What further research could be done? Different statistical analysis?

When working with museum quality pieces, that need destructive analyses performed to

Data obtained for the (NAME TABLE) containing the main component, trace element concentrations and Cu/Zn weight ratio was obtained by ICP-OES and ICP-MS. Collecting samples from 38 ancient ingots to analysis the association between the ingots elemental composition.

We also took a preliminary look at the Copper (Cu), Zinc (Zn), Lead (Pb), and Tin (Sn) as they are the four alloying elements that are known to define orichalcum and historically have been the primary elemental components used to date and pinpoint origin of other orichalcum artifacts.

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